

Mathematics 1

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de Morgan's Laws

August de Morgan (1806-1871): British mathematician, contributed to the development of set theory and Boolean algebra

Formulated some laws as shown below

de Morgan's Law: $\neg (p \vee q) \leftrightarrow \neg p \wedge \neg q$

p	q	$\neg (p \vee q)$	$\neg p \wedge \neg q$	$\neg (p \vee q) \leftrightarrow \neg p \wedge \neg q$
T	T	F	F	T
T	F	F	F	T
F	T	F	F	T
F	F	T	T	T

de Morgan's Laws

de Morgan's Law: $\neg (p \wedge q) \leftrightarrow \neg p \vee \neg q$

p	q	$\neg (p \wedge q)$	$\neg p \vee \neg q$	$\neg (p \wedge q) \leftrightarrow \neg p \vee \neg q$
		NOT (p AND q)	NOT p OR NOT q	
T	T	F	F	T
T	F	T	T	T
F	T	T	T	T
F	F	T	T	T

A copy of the Excel code is given in the Excel file

Truth Tables – Implementation in Excel

p	q	$p \wedge q$	$p \wedge q$ code	$p \vee q$	$p \vee q$ code	$p \oplus q$	$p \oplus q$ code	$\neg p$	$\neg p$ code	$p \rightarrow q$	$p \rightarrow q$ code	$p \leftrightarrow q$
1	1	1	$=p*q$	1	$=p+q-p*q$	0	$=p+q-p*q$	0	$=1-p$	1	$1-p*(1-q)$	1
1	0	0		1		1		0		0		0
0	1	0		1		1		1		1		1
0	0	0		0		0		1		1		1

Translation:

$$T = 1$$

$$F = 0$$

Obvious Statements

Obvious statements: $p \vee T \leftrightarrow T$

Similarly $p \wedge F \leftrightarrow F$

$p \vee T \leftrightarrow T$

p	$p \vee T$	$p \vee T \leftrightarrow T$
T	T	T
F	T	T

$p \wedge F \leftrightarrow F$

p	$p \wedge F$	$p \wedge F \leftrightarrow F$
T	F	T
F	F	T

Simplifications of Truth Tables

$$p \vee F \leftrightarrow p$$

p	$p \vee F$	$p \vee F \leftrightarrow p$
T	T	T
F	F	T

$$p \wedge T \leftrightarrow p$$

p	$p \wedge T$	$p \wedge T \leftrightarrow p$
T	T	T
F	F	T

Excluded Middle

A condition known as an *excluded middle* arises for $p \vee \neg p$

It is evident that

$$p \vee \neg p \leftrightarrow T$$

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Contradiction

A contradiction links two statements, such as p and $\neg p$,

$$p \wedge \neg p$$

It is evident that

$$p \wedge \neg p \leftrightarrow F$$

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Double Negation

Two negatives make a positive (in multiplication). Thus

$$\neg(\neg p) \leftrightarrow p$$

p	$\neg p$	$\neg(\neg p)$
T	F	T
F	T	F

Implication Truth Table

The implication truth table has been covered before. It is noted that implication, $p \rightarrow q$, may also be expressed as

$$p \rightarrow q \iff \neg p \vee q$$

See table below:

p	q	$p \rightarrow q$	$\neg p \vee q$	$p \rightarrow q \iff \neg p \vee q$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

Truth Table for Equivalence

The equivalence truth table has been covered before. It is noted that

$$(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

See table below:

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$A \equiv (p \rightarrow q) \wedge (q \rightarrow p)$	$(p \leftrightarrow q)$
T	T	T	T	T	T	T
T	F	F	F	T	F	F
F	T	F	T	F	F	F
F	F	T	T	T	T	T

Exportation

With exportation the following equivalence is referred to

$$(p \wedge q) \rightarrow r \iff p \rightarrow (q \rightarrow r)$$

p	q	r	$p \wedge q$	$A \equiv (p \wedge q) \rightarrow r$	$q \rightarrow r$	$B \equiv p \rightarrow (q \rightarrow r)$	$A \leftrightarrow B$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	F	T	T	T	T
F	T	F	F	T	F	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

Contrapositive

Contrapositive $\Leftrightarrow$ modus tollens

Negative form of $p \rightarrow q$ is $\neg q \rightarrow \neg p$

Note the change in order of p

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$	$p \rightarrow q \leftrightarrow \neg q \rightarrow \neg p$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

Reductio ad absurdum

Reductio ad absurdum $\Leftrightarrow$ reduction to absurdity

This argument starts with p implies q and also p implies $\neg q$
This is absurd.

$$(p \rightarrow q) \wedge (p \rightarrow \neg q) \Leftrightarrow \neg p$$

p	q	$p \rightarrow q$	$p \rightarrow \neg q$	$(p \rightarrow q) \wedge (p \rightarrow \neg q)$	$(p \rightarrow q) \wedge (p \rightarrow \neg p) \Leftrightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	T
F	T	T	T	T	T
F	F	T	T	T	T

Reductio ad absurdum (negative)

Negative formulation: $(\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q) \leftrightarrow p$

p	q	$\neg p \rightarrow q$	$\neg p \rightarrow \neg q$	$(\neg p \rightarrow q) \wedge$ $(\neg p \rightarrow \neg q)$	$(\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$ $\leftrightarrow p$
T	T	T	T	T	T
T	F	T	T	T	T
F	T	T	F	F	T
F	F	F	T	F	T

Modus Ponens

Modus ponens $\Leftrightarrow$ affirming mode

argument in 3 parts: major premise, minor premise, conclusion

1. If an integer is positive, it is a natural number. *major premise*
2. 25 is a positive integer *minor premise*
3. 25 is a natural number *conclusion*

Let p be the positive integer

Let q be a natural number

$\Rightarrow$

- major proposition $p \rightarrow q$
- minor proposition p

This makes

$$(p \rightarrow q) \wedge p \rightarrow q$$

Modus Ponens

$$(p \rightarrow q) \wedge p \rightarrow q$$

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$(p \rightarrow q) \wedge p \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Proof of Case

$$[(q \vee p) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$$

p	q	r	$q \vee p$	$p \rightarrow r$	$q \rightarrow r$	$\mathbf{A} \equiv \left[\begin{array}{l} (q \vee p) \wedge \\ (p \rightarrow r) \wedge \\ (q \rightarrow r) \end{array} \right]$	$\mathbf{A} \rightarrow \mathbf{r}$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

Summary of Logical Identities

	Identity	Law
1	$\neg(\neg p)$	Double negation
2	$p \vee T \leftrightarrow T$	Simplification
3	$p \wedge F \leftrightarrow F$	Simplification
4	$p \wedge T \leftrightarrow p$	Simplification
5	$p \vee F \leftrightarrow p$	Simplification
6	$p \vee \neg p \leftrightarrow T$	Excluded middle
7	$p \wedge \neg p \leftrightarrow F$	Contradiction
8	$p \vee p \leftrightarrow p$	Idempotenz of $\vee$
9	$p \wedge p \leftrightarrow p$	Idempotenz of $\wedge$
10	$p \vee q \leftrightarrow q \vee p$	Commutivity of $\vee$
11	$p \wedge q \leftrightarrow q \wedge p$	Commutivity of $\wedge$

Summary of Logical Identities

	Identity	Law
12	$(p \vee q) \vee r \leftrightarrow p \vee (q \vee r)$	Associativity of $\vee$
13	$(p \wedge q) \wedge r \leftrightarrow p \wedge (q \wedge r)$	Associativity of $\wedge$
14	$\neg(p \vee q) \leftrightarrow \neg p \wedge \neg q$	de Morgan's Law
15	$\neg(p \wedge q) \leftrightarrow \neg p \vee \neg q$	de Morgan's Law
16	$p \wedge (q \vee r) \leftrightarrow (p \wedge q) \vee (p \wedge r)$	Distributivity of $\wedge$ over $\vee$
17	$p \vee (q \wedge r) \leftrightarrow (p \vee q) \wedge (p \vee r)$	Distributivity $\vee$ over $\wedge$
18	$p \rightarrow q \leftrightarrow \neg p \vee q$	Implication
19	$(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$	Equivalence
20	$(p \wedge q) \rightarrow r \leftrightarrow p \rightarrow (q \rightarrow r)$	Exportation
21	$(p \rightarrow q) \leftrightarrow \neg q \rightarrow \neg p$	Contrapositive
22	$(p \rightarrow q) \wedge (p \rightarrow \neg q) \rightarrow \neg p$	Reductio ad absurdum

	Identity	Law
23	$(\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q) \rightarrow p$	Reductio ad absurdum
24	$(p \rightarrow q) \wedge p \rightarrow q$	Modus ponens
25	$(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r) \rightarrow r$	Proof of cases

Set Theory

Empty set: is a set with no members, symbols: $\emptyset$ or $\{ \}$

Set S with $S = \{a, b, d, f\}$.

Then

$a \in S, b \in S, d \in S, f \in S$ are elements of S .

The set contains 4 elements, denoted as $[S] = 4$. The number of elements is called the **cardinality** of S .

A set may contain other sets, e.g.

$$T = \{\{a, b\}, c, \{d, e, f\}\}$$

Given

$$T = \{\{a, b\}, c, \{d, e, f\}\}$$

Determine the cardinality of T .

Determine the elements of T .

Answer:

$$[T] = 3$$

Elements of T :

$$\{a, b\} \in T, \quad c \in T, \quad \{d, e, f\} \in T$$

Subsets, Supersets and the Universal Set

Consider the set of natural numbers, denoted by $\mathbb{N}$. This set may be divided into various subsets.

Example:

Let E be the set of all even numbers: $E = \{2, 4, 6, 8, \dots\}$

Let O be the set of all odd numbers: $O = \{1, 3, 5, 7, \dots\}$

Clearly E and O are subsets of $\mathbb{N}$. This is written as

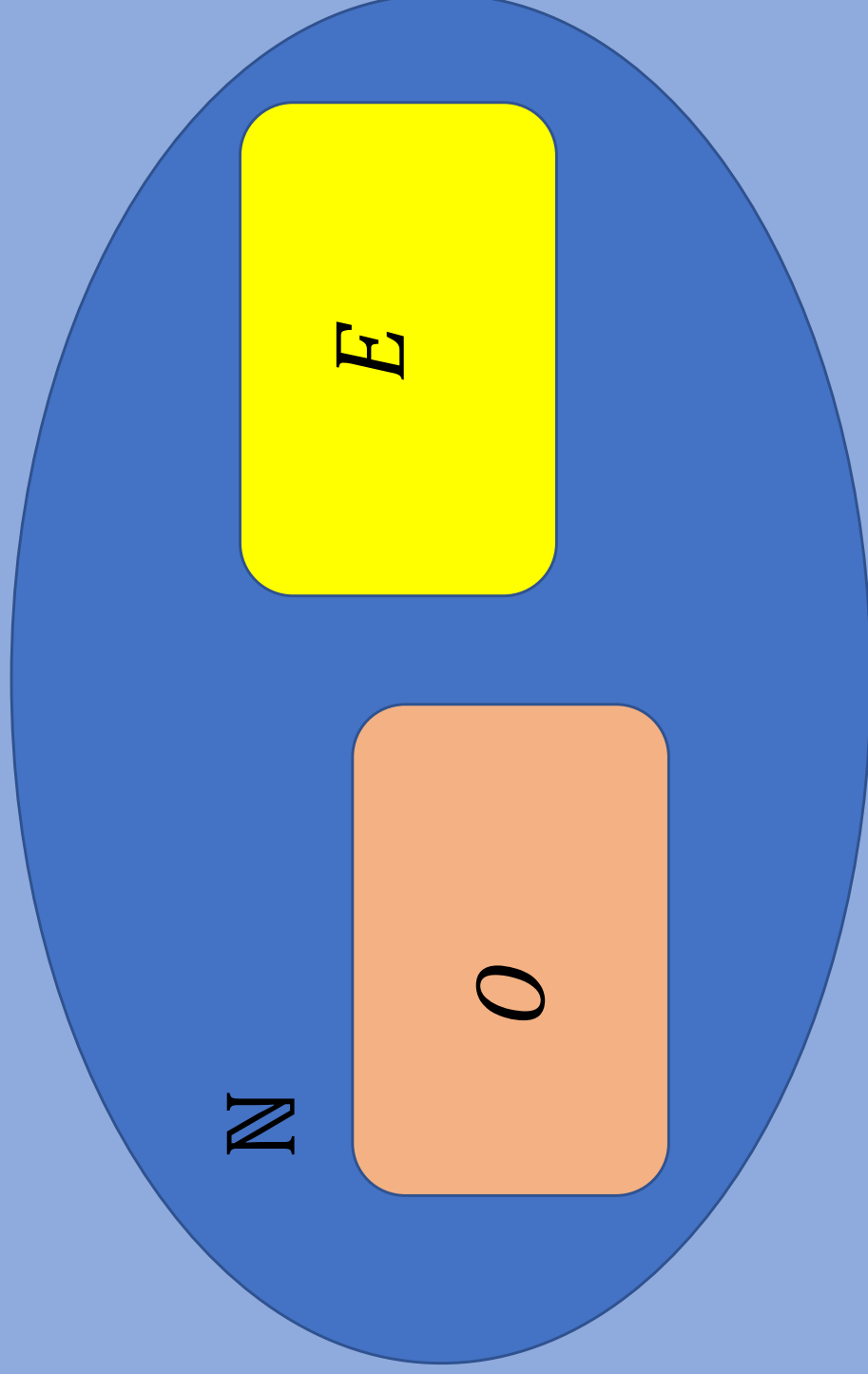
$$E \subset \mathbb{N} \text{ and } O \subset \mathbb{N}$$

In reverse, $\mathbb{N}$ is a superset of E and O . This is written as

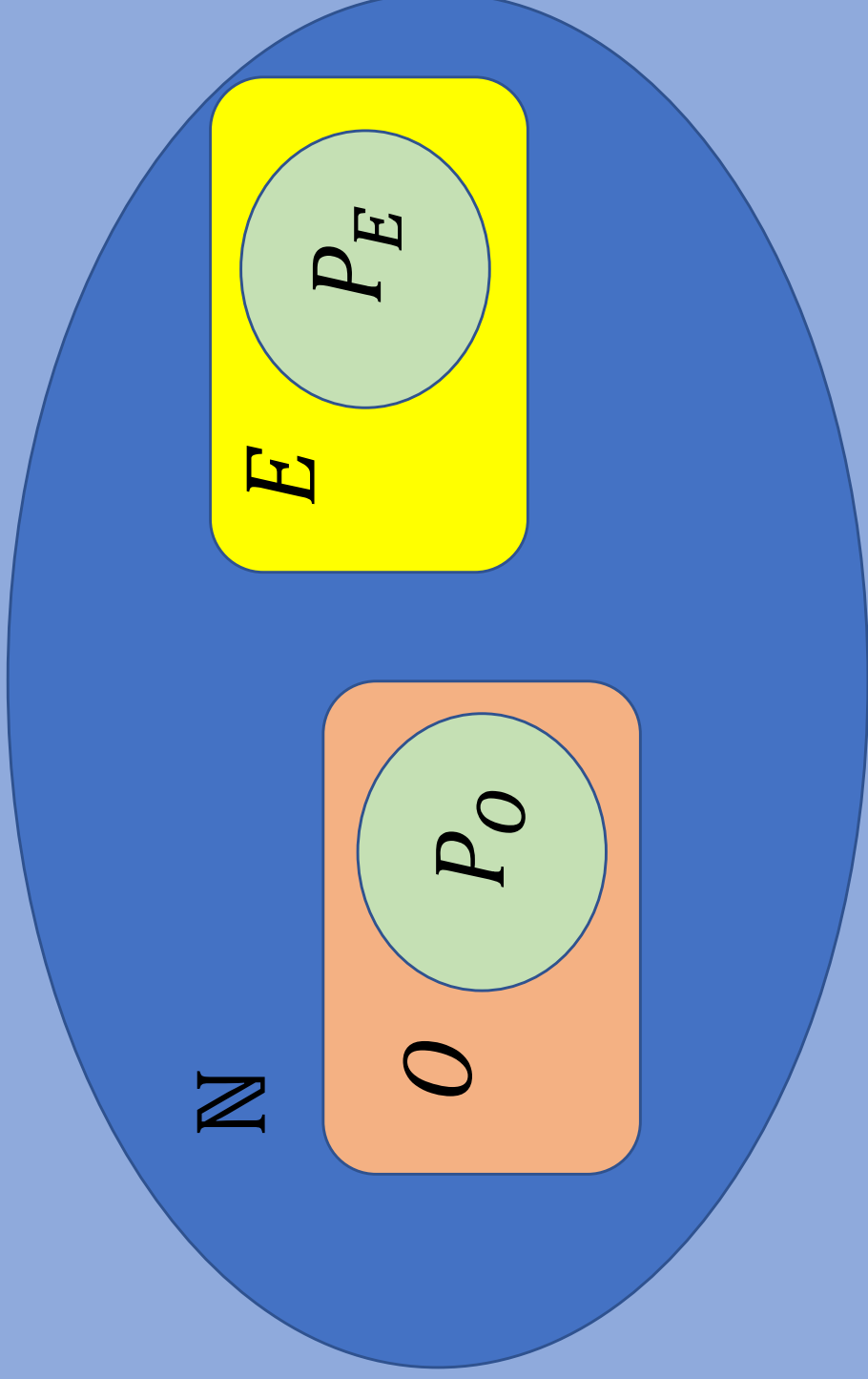
$$\mathbb{N} \supset E \text{ and } \mathbb{N} \supset O$$

Venn Diagrams

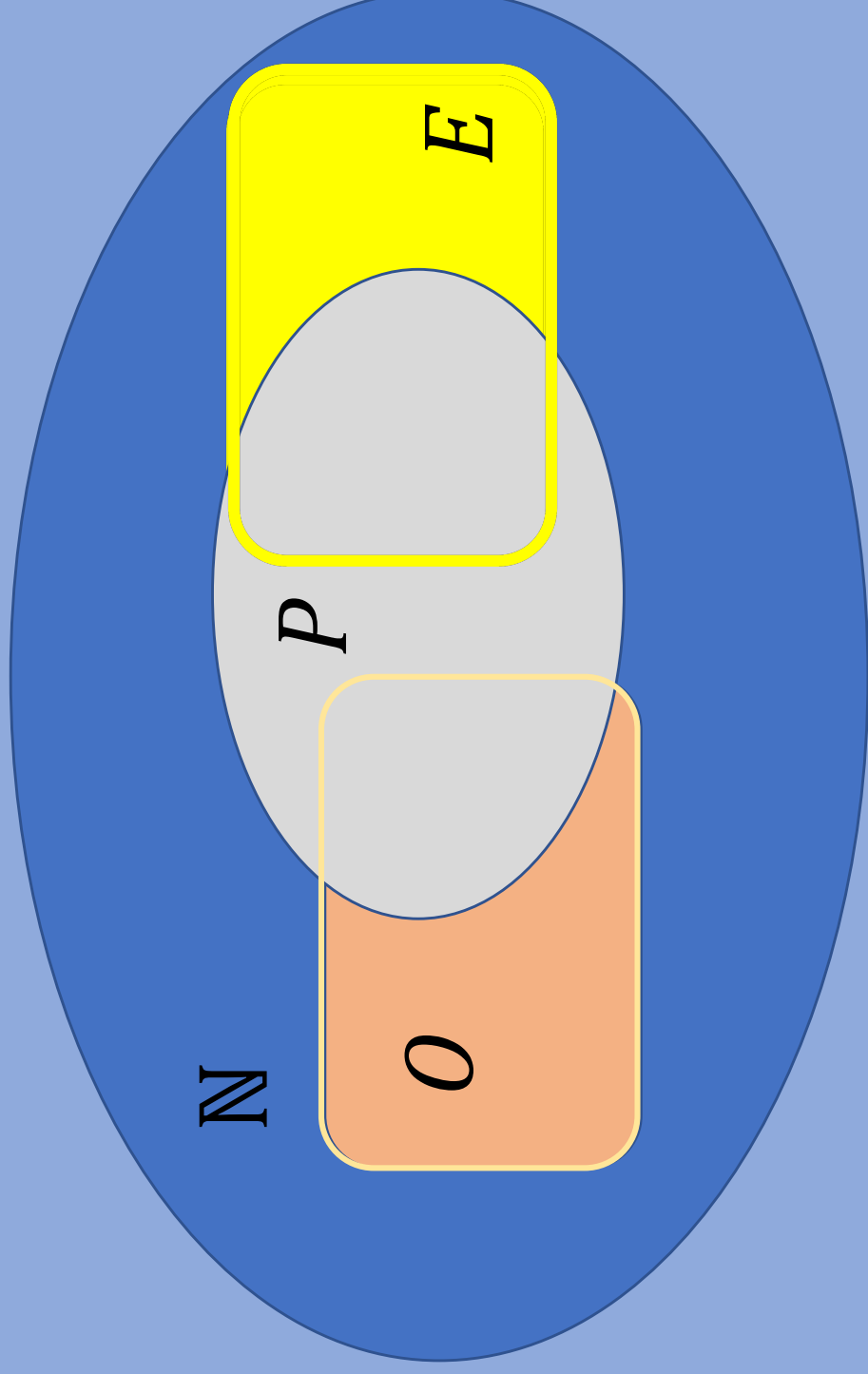
The relationship may also be depicted graphically.



Let P_E be the set of all even prime numbers.
Let P_O be the set of all odd prime numbers.
Then the picture is changed accordingly



The set of all prime numbers P has as subset both P_E as well P_O . The picture is changed to



Set building

$$S = \{n | n \in \mathbb{N} \wedge 1 \leq n \leq 5\} = \{1, 2, 3, 4, 5\}$$

Read: set of all elements n for which n is an element of $\mathbb{N}$ and n is smaller or equal to x and smaller or equal to 5 .

$$S = \{n | \exists n \in \mathbb{N} \quad n = 5m, m \in \mathbb{N}\} = \{5, 10, 15, 20, \dots\}$$

Read: set of all elements n for which there is an n element which is equal to an integer m times 5 .

$$S = \{n | \exists n \in \mathbb{N} \quad n = m^2\} = \{1, 4, 9, 16, \dots\}$$

Read: set of all elements n for which there is an n element m which is equal to the square of an integer m .

Unions of sets

$$S = \{1, 2, 3, 4, 5\}$$

and

$$T = \{1, 3, 4, 7, 9\}$$

Then the union of S and T is

$$S \cup T = \{1, 2, 3, 4, 5, 7, 9\}$$

Note: repeated elements only appear once.

Then the intersection of S and T is

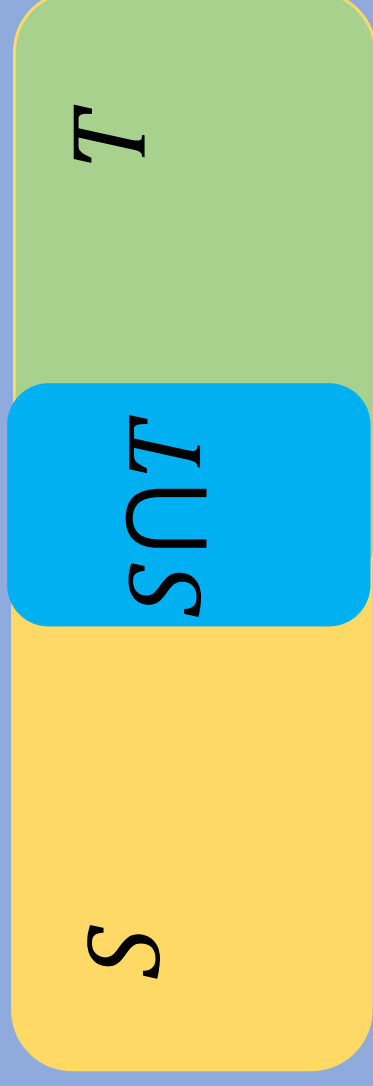
$$S \cap T = \{1, 3, 4\}$$

Unions of sets - graphics

$S = \{1, 2, 3, 4, 5\}$ and $T = \{1, 3, 4, 7, 9\}$

$S \cup T = \{1, 2, 3, 4, 5, 7, 9\}$

$S \cap T = \{1, 3, 4\}$



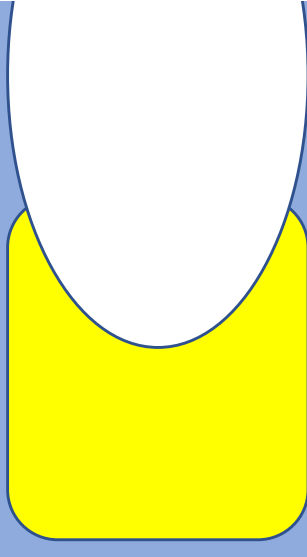
Relative Complement

The relative complement between two sets is the set elements belonging to one, but not the other.

For $S = \{1, 2, 3, 4, 5\}$ and $T = \{1, 3, 4, 7, 9\}$

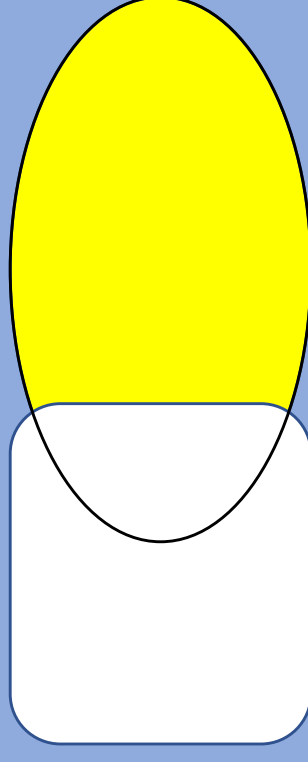
$$S \setminus T = \{2, 5\}$$

Read: S without T, or: S but not T



Determine $T \setminus S$

Answer: $T \setminus S = \{7, 9\}$



Absolute Complement

The absolute complement of a set A is the difference between the universal set U and A . This is written as

$$A^c = U \setminus A$$

Example: Let $U = \mathbb{N}$ and $A = \{n | n \in \mathbb{N} \wedge n \geq 5\}$

Then $A^c = U \setminus A = \{1, 2, 3, 4\}$

Power Set

Given a set S . The power set $\wp(S)$ is the set of all subset of S .
The power set includes the empty set $\emptyset$ and S itself.
Therefore for

$$S = \{1, 2, 3, 4\}$$

Have

$$\wp(S) = \left\{ \emptyset, 1, 2, 3, 4, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{2, 3, 4\}, \{1, 3, 4\}, \{1, 2, 4\}, \{1, 2, 3\}, \{1, 2, 3, 4\} \right\}$$

Example

Given a set S

$$S = \{a, b, c\}$$

Determine the power set $\wp(S)$.

Answer:

$$S = \{a, b, c\}$$

power set $\wp(S)$.

$$\wp(S) = \{\emptyset, a, b, c, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

Given a set S with n elements. Determine the cardinality of $\wp(S)$.

$$\text{Answer: } |S| = n \text{ then } |\wp(S)| = 2^n$$